

## No.of Functional Features Included in the Solution

|                     |                                                                 |
|---------------------|-----------------------------------------------------------------|
| <b>Date</b>         | 15March2024                                                     |
| <b>TeamID</b>       | XXXXXX                                                          |
| <b>ProjectName</b>  | Smart Lender-Applicant Credibility Prediction for Loan Approval |
| <b>MaximumMarks</b> | 5 Marks                                                         |

### FunctionalFeaturesOverview:

This document lists all the functional features included in the project solution . Each feature is described with its purpose, implementation status, and contribution to the overall solution.

| S.No | FeatureName        | FeatureDescription                    | Module / Component | Status (Done/ In Progress / Pending) | Marks Contribution |
|------|--------------------|---------------------------------------|--------------------|--------------------------------------|--------------------|
| 1    | User Input Form    | Collects applicant details            | Flask U            | Done                                 | High               |
| 2    | Data Preprocessing | Cleans and prepares input data        | Python, Pandas     | Done                                 | High               |
| 3    | Loan Prediction    | Predicts loan approval using XGBoost  | ML Model           | Done                                 | High               |
| 4    | Model Selection    | Uses the best-performingtrained model | XGBoost            | Done                                 | Medium             |
| 5    | Prediction Result  | Displays loan approval/rejection      | Flask              | Done                                 | High               |
| 6    | Input Validation   | Checks user input before prediction   | Flask              | Done                                 | Medium             |
| 7    | Model Integration  | Connects ML model with web applicatio | Flask + Python     | Done                                 | High               |
| 8    | Deployment Support | Ready for IBM Cloud deployment        | IBM Cloud          | Done                                 | Medium             |

### FeatureSummary:

| Metric                   | Count/Value |
|--------------------------|-------------|
| TotalFeaturesPlanned     | 8           |
| TotalFeaturesImplemented | 8           |

|                                 |                                                                                              |
|---------------------------------|----------------------------------------------------------------------------------------------|
| Core/Must-HaveFeatures          | Login, LoanApplicationForm, Data Preprocessing, ModelTraining, LoanPrediction, ResultDisplay |
| Additional/Nice-to-HaveFeatures | FlaskWebIntegration, IBMCloud Deployment                                                     |
| FeaturesTested&Verified         | All8featurestestedandworkingsuccessfully                                                     |

**FeatureCategoryBreakdown:**

| S.No | Category                 | FeaturesinCategory | ExampleFeatures                                         |
|------|--------------------------|--------------------|---------------------------------------------------------|
| 1    | UserInterface(UI)        | 2                  | Login Page, LoanApplicationForm                         |
| 2    | Backend/Logic            | 3                  | DataPreprocessing, MLModelPrediction, Result Generation |
| 3    | Database/Storage         | 1                  | Applicant Data Storage                                  |
| 4    | API/ Integration         | 1                  | FlaskIntegrationwithXGBoostModel                        |
| 5    | Security/ Authentication | 1                  | User Login Authentication                               |